

PHYS 210 | Homework 5 · Electric potential and energy

Solution key and walkthrough

Follow the energy: distinguish potential V from energy U , keep charge signs, and use initial minus final energy for electric work.

Due October 13, 2026, 11:59 p.m. (saved course schedule; confirm changes in D2L)

Use $k = 8.988 \times 10^9 \text{ N m}^2/\text{C}^2$, $e = 1.602 \times 10^{-19} \text{ C}$, $m_p = 1.673 \times 10^{-27} \text{ kg}$, and $m_e = 9.109 \times 10^{-31} \text{ kg}$. Convert μC to 10^{-6} C , nC to 10^{-9} C , and grams to kilograms.

23.1 - Work along a two-dimensional move

-0.356 J .

23.5 - Closest approach of repelling charges

12.5 m/s ; minimum separation 0.323 m .

23.7 - Two protons approaching head-on

Maximum force $1.94 \times 10^{-5} \text{ N}$.

23.8 - Energy of three equal charges

0.0971 J .

23.19 - Potentials at two points

$V_a = -737 \text{ V}$; $V_b = -704 \text{ V}$; $W = +8.24 \times 10^{-8} \text{ J}$.

23.27 - Electron on a charged ring axis

Oscillatory motion; centre speed $1.67 \times 10^7 \text{ m/s}$.

23.1 - Work along a two-dimensional move

Given / find: Fixed $q_1 = +2.40 \mu\text{C}$ at the origin; $q_2 = -4.30 \mu\text{C}$ moves from $(0.150, 0) \text{ m}$ to $(0.250, 0.250) \text{ m}$. Find work done by the electric force.

Why this method: Electric work equals initial minus final potential energy.

1. Measure distances from the fixed charge

The final distance is not just the x coordinate. Use Pythagoras from the origin.

$$r_i = 0.150 \text{ m}, \quad r_f = \sqrt{0.250^2 + 0.250^2} = 0.353553 \text{ m}$$

2. Write initial minus final potential energy

Use $U = kq_1q_2/r$. Electric work is $W = U_i - U_f$. Keep the negative sign of q_2 .

$$W = kq_1q_2\left(\frac{1}{r_i} - \frac{1}{r_f}\right)$$

3. Substitute in SI units

μC means 10^{-6} C . The distance bracket is $6.6667 - 2.8284 = 3.8382 \text{ m}^{-1}$.

$$W = (8.988 \times 10^9)(2.40 \times 10^{-6})(-4.30 \times 10^{-6})(3.8382) = -0.356 \text{ J}$$

Answer: -0.356 J .

Check: Moving unlike charges apart raises potential energy; electric work is negative.

Common mistake: Dropping charge signs, or calculating work by an external agent instead of work by the electric force.

23.5 - Closest approach of repelling charges

Given / find: Fixed $q_1 = -2.80 \mu\text{C}$; moving $q_2 = -7.80 \mu\text{C}$, $m = 1.50 \text{ g}$, initially 0.800 m away at 22.0 m/s toward q_1 . Find speed at 0.400 m and minimum separation.

Why this method: Repulsion converts kinetic energy into electric potential energy.

1. Find the constant product kq_1q_2

Both charges are negative, so their product and potential energy are positive.

$$A = kq_1q_2 = 0.196298 \text{ J m}$$

2. Find the total starting energy

Convert 1.50 g to 0.00150 kg . Add kinetic and electric potential energy.

$$E = \frac{1}{2}(0.00150)(22.0)^2 + \frac{0.196298}{0.800} = 0.608372 \text{ J}$$

3. Find kinetic energy at 0.400 m

Potential energy increases as the repelling charges get closer. Subtract it from the same total energy.

$$K = 0.608372 - \frac{0.196298}{0.400} = 0.117628 \text{ J}$$

4. Convert kinetic energy to speed

Rearrange $K = \frac{1}{2}mv^2$.

$$v = \sqrt{\frac{2K}{m}} = 12.5 \text{ m/s}$$

5. Set kinetic energy to zero at the turning point

The moving sphere stops momentarily at minimum separation. All the mechanical energy is then electric potential energy.

$$r_{\min} = \frac{A}{E} = 0.323 \text{ m}$$

Answer: 12.5 m/s ; minimum separation 0.323 m .

Check: The turning point is closer than 0.400 m , so the intermediate speed is real and smaller than 22 m/s .

Common mistake: Setting potential energy to zero at closest approach. It is kinetic energy that vanishes there.

23.7 - Two protons approaching head-on

Given / find: Two protons start far apart, each moving at 2.00×10^5 m/s toward the other in the Earth frame; $m_p = 1.673 \times 10^{-27}$ kg. Find their maximum repulsive force.

Why this method: Both protons contribute kinetic energy, and both stop momentarily at closest approach.

1. Include both moving protons

Their equal and opposite momenta make the centre of mass stationary. Both stop momentarily at closest approach. Initial potential energy at several metres is negligible.

$$K_i = 2\left(\frac{1}{2}m_p v^2\right) = m_p v^2 = 6.692 \times 10^{-17} \text{ J}$$

2. Turn all initial kinetic energy into pair energy

There is one interacting pair. Set $k e^2/r_{\min}$ equal to the total kinetic energy.

$$r_{\min} = \frac{k e^2}{K_i} = 3.45 \times 10^{-12} \text{ m}$$

3. Use Coulomb's law at the smallest separation

Force is largest when r is smallest. Use the unrounded separation.

$$F_{\max} = \frac{k e^2}{r_{\min}^2} = 1.94 \times 10^{-5} \text{ N}$$

Answer: Maximum force 1.94×10^{-5} N.

Check: Using only one proton's kinetic energy would give twice the separation and one-quarter the force.

Common mistake: Using only one proton's kinetic energy, or putting the relative speed 4.00×10^5 m/s into the kinetic energy of each proton.

23.8 - Energy of three equal charges

Given / find: Three charges $q = +1.20 \mu\text{C}$ occupy an equilateral triangle of side $a = 0.400 \text{ m}$. Find the energy relative to infinite separation.

Why this method: Count distinct pairs once: there are three.

1. List each unique pair

The three pairs are (1,2), (1,3), and (2,3). Every distance is 0.400 m . Do not count both (1,2) and (2,1).

2. Find energy per pair

Use $q = 1.20 \times 10^{-6} \text{ C}$ and $U_{\text{pair}} = kq^2/a$.

$$U_{\text{pair}} = \frac{(8.988 \times 10^9)(1.20 \times 10^{-6})^2}{0.400} = 0.0323568 \text{ J}$$

3. Add three identical contributions

All charges have the same sign, so every contribution is positive.

$$U = 3U_{\text{pair}} = 0.0970704 \text{ J}$$

Answer: 0.0971 J .

Check: The energy is positive: external work is required to assemble repelling charges.

Common mistake: Double-counting the pairs. The system has three pair energies, not six.

23.19 - Potentials at two points

Given / find: $q_1 = +2.40 \text{ nC}$ and $q_2 = -6.50 \text{ nC}$ are 0.100 m apart. A is their midpoint; B is 0.0800 m from q_1 and 0.0600 m from q_2 . Find V_a , V_β , and electric work moving $+2.50 \text{ nC}$ from B to A.

Why this method: Potential is a signed scalar sum; work uses start minus finish.

1. Find the potential at A

A is halfway along the 0.100 m separation, so it is 0.0500 m from each charge. Potential adds as signed scalars, without vector components.

$$V_A = k \left(\frac{2.40 \times 10^{-9}}{0.0500} - \frac{6.50 \times 10^{-9}}{0.0500} \right) = -737.016 \text{ V}$$

2. Find the potential at B

Use the different stated distances: 0.0800 m to q_1 and 0.0600 m to q_2 .

$$V_B = k \left(\frac{2.40 \times 10^{-9}}{0.0800} - \frac{6.50 \times 10^{-9}}{0.0600} \right) = -704.060 \text{ V}$$

3. Take starting minus ending potential

The test charge goes from B to A, so use $V_B - V_A$. Multiply by $+2.50 \text{ nC}$.

$$W_{B \rightarrow A} = (2.50 \times 10^{-9})(-704.060 + 737.016) = 8.24 \times 10^{-8} \text{ J}$$

Answer: $V_a = -737 \text{ V}$; $V_\beta = -704 \text{ V}$; $W = +8.24 \times 10^{-8} \text{ J}$.

Check: The positive test charge moves to lower potential, so the field does positive work.

Common mistake: Reversing $V_B - V_A$ or rounding both potentials too early. Keep guard digits in the small difference.

23.27 - Electron on a charged ring axis

Given / find: Ring radius $R = 0.150$ m, charge $Q = +24.0$ nC. An electron starts at rest on its axis, $x_0 = 0.300$ m; constrain motion to the axis. Describe motion and find the speed through the centre.

Why this method: The attractive force restores the electron toward the ring plane.

1. Write the ring's on-axis potential

Every small ring element is the same distance $\sqrt{R^2 + x^2}$ from a point on the axis. Add their scalar potentials.

$$V(x) = \frac{kQ}{\sqrt{R^2 + x^2}}$$

2. Compare the release point with the centre

Use $R = 0.150$ m, $x_0 = 0.300$ m and $Q = 24.0 \times 10^{-9}$ C.

$$V(0) = 1438.08 \text{ V}, \quad V(x_0) \approx 643.13 \text{ V}$$

3. Use the electron's negative charge

$U = -eV$ decreases as V increases. Starting from rest, the lost potential energy becomes kinetic energy.

$$\frac{1}{2}m_e v^2 = e[V(0) - V(x_0)]$$

4. Solve for the centre speed

The potential rise is about 794.95 V.

$$v = \sqrt{\frac{2(1.602 \times 10^{-19})(794.95)}{9.109 \times 10^{-31}}} = 1.67 \times 10^7 \text{ m/s}$$

5. Describe what happens after crossing the centre

The force is zero exactly at the centre, but the electron has its maximum speed there. It continues through, slows down, stops at $x = -0.300$ m, and returns. Neglecting radiation and losses, it oscillates between ± 0.300 m. This amplitude is not in the small-displacement simple-harmonic regime.

Answer: Oscillatory motion; centre speed 1.67×10^7 m/s.

Check: It oscillates between ± 0.300 m, fastest at the centre. This finite-amplitude motion is not simple harmonic because force is not proportional to x .

Common mistake: Thinking zero force at the centre means zero speed. The speed is greatest there; inertia carries the electron across.

Source: course handout linked from the online walkthrough. Independent study key, not an instructor-issued answer key.